

Labor and Product Market Effects of Mergers*

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Abstract

We use a two-level vertical supply chain model to forecast the labor and product market impacts of mergers. In the model, firms and workers simultaneously negotiate wages while firms engage in Bertrand price competition. Merger simulations conducted with U.S. hospital and Colombian manufacturing data illustrate how mergers can reduce worker welfare by reducing workers' bargaining leverage and/or increasing firms' bargaining leverage. Worker harm is highly correlated with the merger-induced increase in concentration. Finally, we find that conventional merger simulation screens only considering downstream competition can frequently identify the mergers that harm workers when firms compete in the product market.

Keywords: bargaining models; merger simulation; vertical markets; labor market structures

JEL classification: L13; L40; L41; J42

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1 Introduction

Over the last forty years, industrial organization economists have developed models to forecast the product market effects of horizontal and vertical mergers. These tools, including differentiated Bertrand models for posted prices (Werden and Froeb, 1994) and Nash Bargaining for negotiated prices (Gowrisankaran et al., 2015) describe the price-setting processes in relevant markets. However, growing evidence suggests that mergers can also harm workers, especially in labor markets that are not perfectly competitive (Sokolova and Sorensen, 2021). In particular, recent empirical studies suggest that mergers of competing employers in highly concentrated labor markets can significantly reduce wages (Prager and Schmitt, 2021b; Arnold, 2021). In response, scholars have urged antitrust authorities to consider how horizontal mergers affect workers in addition to consumers (Naidu et al., 2018; Marinescu and Hovenkamp, 2019; Card, 2022). Yet, existing methodological approaches do not jointly model wage negotiations between workers and employers alongside strategic interaction in the product market using inputs available to antitrust authorities.¹

In this paper, we adapt models frequently used to study the competitive impact of horizontal and vertical mergers on the product market to simulate the labor market effects of mergers. We use a two-level vertical supply chain to examine how a merger of producers affects both competition in the final goods market and the welfare of production workers. Our model, which is analytically similar to that used by Sheu and Taragin (2021), assumes that upstream workers collectively bargain over linear wages with downstream manufacturers who employ Leontief production technology to manufacture differentiated products. This specification generates constant marginal costs, ensuring our framework remains consistent with the standard cost assumptions used in virtually all antitrust merger simulations. In particular, our model allows mergers to affect the bargaining leverage of both the merging

¹Berger et al. (2023) have recently developed a merger simulation technique related to Berger et al. (2022) to forecast the effects of mergers on workers, but they do not include strategic interaction in the product market in their simulations. For a review of the economic literature as it relates to antitrust enforcement in labor markets, please see Prager (2025).

parties and workers. To close the model, we assume that downstream manufacturers sell differentiated products where prices are determined via a Bertrand game with logit demand (Werden and Froeb, 1994).

Our model is designed to analyze mergers in environments where firms earn positive rents and workers may share in those rents. To evaluate how consolidation affects workers, we build on the Nash-in-Nash bargaining framework of Horn and Wolinsky (1988), in which wages divide the surplus generated by the firm–worker match relative to each party’s outside option. We adapt this framework to a merger simulation setting by tying workers’ disagreement payoffs to opportunities in an external competitive labor market. This structure allows us to characterize how a merger’s effects on both the product market and the local labor market feed directly into the surplus workers can negotiate. In particular, eliminating an independent employer through a merger changes workers’ available outside options, thereby shifting the bargaining environment in a way that standard merger simulation models do not capture.

Our approach contrasts with the modern monopsony literature, which models firms as facing upward-sloping labor supply curves arising from search frictions or job differentiation (Burdett and Mortensen, 1998; Manning, 2003). Recent work by Setzler (2025) incorporates hospital-specific labor supply elasticities into a structural model of hospital pricing and quality to study how mergers affect both product- and labor-market outcomes. These models emphasize wage-setting when firms face imperfectly elastic labor supply. Unfortunately such models are difficult to implement in the context of an antitrust investigation because market-specific labor supply elasticity estimates are not readily available. By contrast, our contribution is to provide a tractable merger simulation framework that embeds how consolidation alters the division of rents between firms and workers into widely used models of differentiated product competition.

In our framework, mergers affect worker welfare through four distinct mechanisms, driven by the interplay between product and labor market structure. First, two opposing effects

arise directly from the product market: when merging firms compete in the same product market, the post-merger reduction in output depresses labor demand, harming workers. However, the associated increase in product market rents expands the total surplus available for negotiation, which can put upward pressure on wages. Second, the merger alters the bargaining leverage of both parties. On the firm side, a merger increases leverage if the combined entity can recapture lost sales at an alternative production site should local negotiations fail. Crucially, on the worker side, mergers that reduce the number of independent local employers erode workers' bargaining leverage by eliminating an outside employment option from their threat point. The net effect of a merger on worker welfare is determined by the summation of these four forces.

To illustrate how our model predicts the labor and product market effects of mergers, we conduct simulations across different geographic market configurations for products and labor, using data typically available in merger investigations. These simulations rely on publicly available data: for non-tradable goods, we use U.S. hospital industry data on inpatient medical services; for tradable goods, we use data from the Colombian Manufacturing Census.

To isolate the labor market effects from the product market effects, we focus on three specific merger scenarios defined by the nature of the market overlap. First, we examine mergers where the parties compete in both product and labor markets ($\Delta\text{HHI}_L > 0$ and $\Delta\text{HHI}_P > 0$); in this case, workers are subject to the full suite of mechanisms.² Second, we isolate the labor channel by simulating mergers with overlap solely in the labor market ($\Delta\text{HHI}_L > 0$ and $\Delta\text{HHI}_P = 0$), where the primary driver of harm is the loss of a worker's outside option. Third, we consider mergers with overlap solely in the product market ($\Delta\text{HHI}_L = 0$ and $\Delta\text{HHI}_P > 0$), which allows us to observe the effects of rent expansion and firm leverage in the absence of labor market consolidation.

In the first case, we use U.S. hospital industry data to simulate mergers where the geo-

²The HHI is the Herfindahl index defined as the sum of the squared market shares of market participants. In our model, mergers can change the HHI in both the product market and the labor market. The ΔHHI_L and ΔHHI_P are the change in the Herfindahl index caused by a merger in the labor and product market, respectively.

geographic market for the service produced is the same as the assumed geographic market for labor.³ In these simulations, we assume that the wages of nursing and pharmacy employees are negotiated. Following Prager and Schmitt (2021b), we focus on this combined category because these workers are the most likely to be negatively affected by mergers, and available data does not permit a finer distinction between nursing administration and pharmacy staff.

Next, we consider mergers that directly affect the labor market but not the product market also using data from the hospital industry. In this scenario, we assume that the geographic market for inpatient services is highly localized as in the first case, but that the geographic market for labor is broader, roughly corresponding to a commuting zone rather than the narrow draw area of hospitals. In this case, we simulate mergers between hospitals in different geographic markets for the product market but assume that both hospitals hire nursing and pharmacy workers from a common labor market.

Finally, we examine “export-oriented” mergers that directly affect the product market but not the local labor market. This situation can occur in tradable goods industries, where products are consumed in regions different from where they are produced. Using Colombian manufacturing data, we simulate mergers between firms in the same industry but with production facilities in different parts of Colombia, so the merging firms are not direct competitors in the labor market.⁴ In these simulations, we assume that only skilled workers’ wages can be affected by mergers.⁵

To illustrate the different mechanisms that cause mergers to affect worker welfare, we

³In all of the simulations, we use publicly available data describing market shares, margins, employment levels and wages to calibrate the demand parameters of our model and determine the relative bargaining power of workers to firms. Using the calibrated model, we then simulate the impact of randomly generated mergers on downstream prices and the wages of affected workers using our model. In the simulation study, we examine the set of all possible mergers between two competing firms within the specified geographic markets where we have sufficient data, and where the merger would increase HHI by at least 100 and the pre-merger HHI is at least 1500.

⁴Our model can accommodate tradable goods mergers involving firms with production facilities in the same labor market. We limit the set of mergers in our simulations to isolate a mechanism affecting bargaining in our framework.

⁵Prager and Schmitt (2021b) also find that skilled workers, but not unskilled workers, were affected by hospital mergers, suggesting that skilled workers are more likely to have industry specific human capital. For this reason, we model skilled workers as those most likely to be harmed by mergers in manufacturing industries.

first simulate mergers using our full model and two alternative nested models. We begin with a canonical ‘downstream only’ merger simulation model, which assumes that mergers only directly affect product market and impact workers solely through reduced employment, as in Werden and Froeb (1994). Next, we allow mergers to also affect workers’ negotiated wages, but assume that the threat point of the worker is exogenously determined as in Horn and Wolinsky (1988). Finally, we simulate mergers using the full model, where the merger removes an employment option from the workers’ threat point, further affecting negotiated wage. Our simulations show that the impact of mergers on workers is typically much larger where workers lose an employment option from their threat point. We take this finding as providing suggestive evidence that mergers can significantly harm workers by reducing workers’ leverage through losing a rival employer in the labor market.

We next separately examine the effects of mergers on employment and wages for each of the three cases we study using merger simulations. For hospital mergers that affect only the labor market, we find that mergers only reduce wages. In principle, lower wages could increase labor demand by lowering marginal costs; however, because nursing and pharmacy wages account for only roughly 2% of hospital costs, the resulting pass-through to output is negligible. Moreover, the reduction in wages is strongly negatively associated with the ΔHHI_L . Simulated hospital mergers that affect both the product and labor market are always predicted to lower wages and output. In principle, these mergers could increase worker wages. While employment falls due to the reduction in product market competition, workers may benefit as increased rents are partially passed through as higher wages. In this case, however, the reduction in worker leverage caused by the merger more than offsets any benefit to the workers from the increased rents transferred from consumers.

Our findings from the simulated mergers with the Colombian Manufacturing Census are, perhaps, the most interesting. For mergers with smaller impacts on competition ($\Delta\text{HHI}_P < 1000$) the change in wages is very small and in some cases positive. In this case, where mergers only affect the product market, the increase in rents available to workers effectively

offsets the increase in firm’s bargaining leverage. This leverage arises because, post-merger, a firm can credibly threaten to recapture sales at its newly acquired plant if negotiations fail at the original facility. However, as the mergers become more competitively significant (i.e., ΔHHI_P increases), simulated wages begin to fall substantially.

Our merger simulation results indicate that if wages are determined by a bargaining process like ours, traditional ‘downstream only’ merger screens may underestimate, often severely, the welfare effects of mergers on workers. However, this does not mean that such screens are ineffective at identifying the mergers that harm workers. In our final analysis, we assess the efficacy of a conventional merger screen in identifying mergers that would harm workers. For hospital mergers that directly affect both the product and labor market (i.e. the geographic markets for inpatient services and labor coincide), we find that traditional screens frequently identify the mergers that harm workers. For example, when the geographic market for the labor and product markets are identical, 77% of mergers that harm workers by at least 1% and 99% of mergers that harm workers by 5% are flagged as harmful by the downstream only merger simulation screen. Crucially, however, for mergers characterized by distinct product markets but overlapping labor markets, standard downstream screens are structurally incapable of detecting harm, as they yield zero predicted price effects despite significant potential wage suppression.

The remainder of our paper is organized as follows. Section 2 discusses how our paper relates to the literatures on labor market competition, bargaining, and merger simulation. Section 3 describes our modeling approach. Section 4 gives details about the labor and product market configurations that we analyze and simulation results from calibrations to the US hospital industry and Colombian manufacturing. Section 5 examines the extent to which a downstream only model is effective at identifying mergers that harm workers. Section 6 concludes.

2 Literature Review

Our paper is related to a rapidly growing literature that examines departures from perfect competition in labor markets.⁶ Multiple studies have found that firms can substantially increase the number of employees they hire by raising the offered wage, a finding inconsistent with the infinite firm labor supply elasticity predicted by a model of perfect competition, e.g., Sokolova and Sorensen (2021). Other research finds evidence of rent sharing between workers and firms: firms experiencing profit shocks appear to share some of those rents with employees. If labor markets were perfectly competitive, firms would have no incentive to share rents with their employees. Labor economists have developed a number of models to explain the source of employer power, most prominently those viewing search frictions as the source of employer power (Burdett and Mortensen (1998) and Manning (2003)). These models, however, rarely allow for employers to take into account the behavior of rival employers in setting wages.⁷ Interestingly, a handful of recent empirical papers find that mergers of competing employers in highly concentrated labor markets reduced workers' wages (Prager and Schmitt, 2021b; Arnold, 2021). These results suggest that in concentrated markets employers may behave strategically in setting wages: employers appear to lower their wages when facing fewer competitors in the labor market following mergers.

Our paper is most similar to a research approach that examines how wages are generated in a model of bargaining (see, e.g., Jarosch et al. (2024) and Schubert et al. (2021)). In this framework, the employee/employer match generates rents, and a worker's wages will be a function of both the worker's opportunity wage and a fraction of the rents generated by the match. By contrast, if the labor force were perfectly competitive, any rents the firm earned would not be passed through to workers. A series of empirical papers have found evidence that workers capture a significant share of rent (between 0.02 and 0.29) consistent with the bargaining approach (Card et al., 2018).

⁶See Card (2022) for a recent review.

⁷A notable exception being Berger et al. (2022) who model employers engaged in Cournot competition for labor.

Our paper also adds to (and draws heavily from) the literature on merger simulation, see, Whinston (2007) and Asker and Nocke (2021) for a general description of these techniques. Most merger simulation research has focused on forecasting the impact of mergers in the product markets where the merger takes place, e.g., Werden and Froeb (1994), Nevo (2001). Implicitly, these papers have assumed that mergers in product markets do not have impacts on input markets. Our paper most closely relates to recent research that has extended merger simulation to allow downstream mergers to also impact behavior in upstream markets and vice versa (Sheu and Taragin, 2021).⁸

Berger et al. (2023) and Setzler (2025) take a distinct and complementary approach to simulating the effects of mergers on workers. Berger et al. (2023) examine mergers between firms that each produce a single, undifferentiated product, focusing on how strategic interaction in labor demand—modeled as Cournot competition—interacts with upward-sloping labor supply. While valuable for homogeneous goods markets, this Cournot assumption differs from the differentiated Bertrand and bargaining frameworks typically used by antitrust agencies. Recently, Setzler (2025) extends this framework to allow for heterogeneous workers and products, with Cournot competition both upstream and downstream. By contrast, our framework explicitly models product differentiation using standard Bertrand methods that are both tractable and widely used by antitrust authorities. Our framework also allows us to capture how rents generated by downstream market power are divided between firms and workers via bargaining with minimal additional data requirements.

3 Model

The effect of horizontal mergers in the product market is well known. Absent efficiencies, most models predict that prices rise and market output falls. What is less understood is how horizontal mergers affect the employees of merging firms and workers at competing

⁸In related work, Craig et al. (2021) estimate the impact of hospital mergers on input markets (medical supplies) and find little evidence that hospital mergers significantly reduce hospitals’ costs. They find that acquired hospital experience a small reduction on costs, while acquiring firms costs are essentially unchanged.

firms. Our contribution is to examine the labor market effects of horizontal mergers by simultaneously modeling the effects of the merger on both the product and labor market.

We adopt the bargaining framework of Horn and Wolinsky (1988) that was originally developed to examine how a union and employer negotiate over wages. While unions only represent about 6% of U.S. private sector employment, unionization rates remain high in many OECD countries including Canada (26%), Italy (32%), and the United Kingdom (23%).⁹ Moreover, in the U.S. unions can play an important role in mergers involving industries scrutinized by antitrust authorities, including: supermarkets (Kroger’s/Albertson’s (2024)), airlines (JetBlue/Spirit (2023), US Airways/American (2013), and publishing (Penguin Random House/Simon & Schuster(2022)).¹⁰

We follow Draganska et al. (2010) and Sheu and Taragin (2021) in relaxing two assumptions made by Horn and Wolinsky (1988). First, we allow for variation in the bargaining power parameter, which allows us to distinguish strong unions that are capable of extracting additional employer surplus from weak unions that are unable to increase wages much beyond the worker’s next best employment option. Second, we allow for the workers’ threat point, or opportunity wage, to be set endogenously. Doing so allows us to analyze how downstream mergers affect both wages and product prices in a vertical supply chain.

In addition, we extend Sheu and Taragin (2021) to accommodate certain features of employer/employee bargaining. First, we explicitly allow for employers that may or may not produce competing products in the same geographic region to be viable employment options for workers (through the workers’ threat point). Second, we examine how a merger affects negotiations by removing an independently operated employment option from the local labor market. Following the approach of Jarosch et al. (2024), post-merger we eliminate workers’ ability to obtain employment at another outlet of the merged firm if workers and the firm

⁹The BLS estimates that the private sector unionization rate was 6.1% in 2021. Estimates of unionization rates in Canada, Italy, and the United Kingdom come from the OECD and are available at: <http://www.oecd.org/employment/ictwss-database.htm>.

¹⁰The FTC’s complaint in Kroger’s/Albertson’s alleges that “The combined Kroger and Albertson’s would have more leverage to impose sub-par terms on union grocery workers that slow improvements to wages, worsen benefits, and potentially degrade working conditions.”.

fail to negotiate a wage. That is, in post-merger negotiations, all outlets now owned by the merged firm are removed from the threat point. Together, these two modifications allow us to examine how changes in the set of employment options available to workers resulting from mergers affect worker welfare.

In the following sections, we present product demand (Section 3.1), the Bertrand equilibrium for product prices (Section 3.2), and the bargaining game for wages, including worker and firm threat points (Section 3.3).

3.1 Product Demand

We assume that product demand follows a standard logit demand framework. There is a finite set of products, $n \in N$. Each consumer chooses a single unit of one product, or instead chooses an outside option, which could be a product outside the product market, or not consuming at all. Product market size is denoted as $\mathcal{S}$. Consumer i receives utility $V_n^i = \delta_n - \alpha p_n + \epsilon_{in}$ from choosing good $n \in \{N, 0\}$, where δ_n denotes the value of the good n to all consumers, p_n the price of good n , $-\alpha$ the dis-utility of spending money, and ϵ_{in} the idiosyncratic taste of consumer i for good n , which is not observed by the firm.

We assume that ϵ_{in} is distributed according to the Gumbel distribution. We also normalize utility from the outside option ($n = 0$) to be zero. Therefore, the number of consumers choosing product n is given by $q_n = \mathcal{S}s_n = \mathcal{S} \frac{\exp(V_n)}{1 + \sum_{n \in N} \exp(V_n)}$, where s_n is the share of consumers choosing product n .

3.2 Price Setting

We assume that product prices are set by firms in Bertrand equilibrium. Each firm may produce one or more products, such that $|J| \leq |N|$. Let Z_j denote the set of products produced by firm $j \in J$.

For ease of exposition, we assume a Leontief production function where each product is

produced using a single unit of labor and non-labor inputs.¹¹ Leontief production implies that labor and non-labor inputs are perfect complements, with each unit of product n costing $c_n + w_n$ to produce.¹²

Firm j maximizes profits over all the products that it owns, $h \in Z_j$, solving the following:

$$\max_{\{p_h\}_{h \in Z_j}} \sum_{h \in Z_j} (p_h - w_h - c_h) q_h.$$

Rearranging the firm's first order condition for product n yields the following expression for margin:

$$p_n - w_n - c_n = -\frac{1}{\alpha(1 - \sum_{h \in Z_j} s_h)} \quad (1)$$

The Bertrand equilibrium is the set of prices p_n for $n \in N$ such that Equation 1 is satisfied for all n .

3.3 Wage Setting Through Bargaining

Wages are set using the Nash-in-Nash bargaining concept of Horn and Wolinsky (1988). Below we outline the payoffs and threat points of firms and workers, and the bargaining solution.

3.3.1 Firm Payoff

We assume that each product n produced by the firm is produced at a dedicated production facility, and that bargaining takes place between the workers at that facility and the firm. For each n , workers collectively bargain for a single wage, w_n , to be paid per unit of output produced. If a firm reaches an agreement with the workers in labor market k producing

¹¹A more general Leontief production function over labor (l_n) and non-labor (k_n) inputs assumes that output $q_n = \min\{\frac{l_n}{\gamma_n^l}, \frac{k_n}{\gamma_n^k}\}$, $\gamma_n^l > 0, \gamma_n^k > 0$. While setting $\gamma_n^l = \gamma_n^k = 1$ eases exposition, our model can be rewritten to accommodate the more general version.

¹²While w_n will be allowed to adjust in equilibrium following the bargaining framework, for tractability we assume that other inputs are sold in a competitive market, meaning c_n will be held fixed.

product n , the firm is able to produce product n and receives profit from n of:

$$\pi_n = (p_n^* - w_n^* - c_n)q_n$$

where w_n^* is the wage solving the bargaining game outlined below, and p_n^* solves the Bertrand game outlined above.

3.3.2 Firm Disagreement Payoff

The assumption we make about the firm disagreement payoff is one commonly used in the literature, see, for example, Draganska et al. (2010) or Gowrisankaran et al. (2015). Under our Leontief production technology, if a firm fails to reach an agreement with the workers producing product n , those marginal units cannot be produced. The firm, however, continues to produce any other products that it owns, and captures additional sales from consumers who substitute from the foregone units of n to other products owned by firm j .

The payoff to the multi-product firm failing to reach an agreement is therefore:

$$\sum_{h \in \{Z_j \setminus \{n\}\}} (p_h - w_h - c_h) \frac{s_h}{1 - s_n} s_n \mathcal{S}$$

Note that a single-product firm will receive zero payoff following a failure to reach an agreement with workers.

3.3.3 Worker Payoff

Suppose there are L distinct labor markets where goods in the product market are manufactured and for each labor market $m \in L$, let $\mathfrak{S}^m$ denote the size of the labor market and $N_m \subseteq N$ denote the set of products produced in labor market m . Assume that each product $n \in N$ is produced in exactly one labor market. In addition, let $\mathfrak{L}_m \subset N_m \cup \mathfrak{o}_m$ denote the set of employment opportunities available to workers in a given labor market, where $\mathfrak{o}_m$ is an outside option specific to labor market m whose wage $w_{\mathfrak{o}_m}$ is exogenously determined

and could represent the payoff from not working at all, or working in a job outside the labor market. For each $n \in N$, let $\mathfrak{s}_n^m$ denote the proportion of workers who produce product n in labor market m and collectively bargain over the wage w_n . Under our Leontief production technology, the number of workers who produce n equals the number of units of n produced, meaning that the labor market share must be equal to the product market share up to the relative market sizes: $\mathfrak{s}_n^m \mathfrak{S}^m = s_n \mathcal{S}$.¹³

Workers care about both the wage paid and the output, their payoff from an agreement with the firm equal to $w_n \mathfrak{s}_n^m \mathfrak{S}^m = w_n s_n(\mathbf{p}) \mathcal{S}$. Workers do not directly bargain over $s_n(\mathbf{p})$ but in equilibrium a higher w_n may result in a lower $s_n(\mathbf{p})$ through increased product prices p_n .

3.3.4 Worker Disagreement Payoff

The workers' disagreement payoff takes a significantly different form than the firm's disagreement payoff. Workers are constrained to find employment in the labor market they live in. As a result, in the event employees and employer negotiations break down, the workers' alternative options are limited to employers located in the same geographic area they live in. This area includes other firms in the product market located in the region where the merger is taking place and potentially employers in different product markets that use similar labor.¹⁴ In addition, like Jarosch et al. (2024), we assume that firms will not make multiple job offers to workers. For this reason, it is not possible for workers to get jobs producing a product owned by the same firm following a failure to reach an agreement in collective bargaining.

A failure of a firm and its workforce to reach an agreement has two effects that are relevant for determining the workers' disagreement payoff. First, consumers substitute away

¹³Note that this flexibility in the market sizes $\mathfrak{S}^m$ and $\mathcal{S}$ allows us to model different labor and product market elasticities that may occur in real world applications.

¹⁴In particular, if the geographic market for labor is larger than the geographic market for products, then employers in the same industry but a different region could be potential employers. For example, the merger of two retailers in a town may hire workers from throughout a commuting zone. It is also possible that firms in different product markets hire workers from the same labor market. For example, manufacturers of bread and beverages may hire truck drivers to deliver products to retailers from the same labor pool in a town.

from discontinued product n to other products produced by rival firms. As a result, demand for substitute products increases. This, in turn, increases the demand for workers at those firms producing substitute products. Second, workers who had previously worked at the firms producing product $n \in Z_j$ will only work at other firms (not producing products in Z_j) located in their labor market or the outside employment option. This assumption will tend to make workers worse off following a merger, since post-merger workers lose an employment option (Z_j expands) when the merging firms are located in the same region. We assume that worker diversion is given by $\frac{s_l^m}{1-s_n^m}$ which, because of Leontief production equals $\frac{s_l(\mathbf{p}) \frac{\mathcal{S}}{\Theta^m}}{1-s_n(\mathbf{p}) \frac{\mathcal{S}}{\Theta^m}}$. The workers' threat point is as follows:

$$\sum_{l \in \{\mathcal{L}_m \setminus Z_j\}} w_l \frac{s_l(\mathbf{p}) \frac{\mathcal{S}}{\Theta^m}}{1 - s_n(\mathbf{p}) \frac{\mathcal{S}}{\Theta^m}} s_n(\mathbf{p}) \mathcal{S}$$

The workers' threat point is increasing in the wages at other firms, and the attractiveness of substitute products to consumers.

3.3.5 Bargaining Solution

Wages are assumed to be set in a Nash bargaining game. Worker surplus is given by the worker payoff from reaching an agreement minus the disagreement payoff, and likewise for firm surplus. The bargaining solution will maximize the product of worker and firm surplus, weighted by bargaining power, which is given by λ_m for each market $m \in M$. The bargaining problem is therefore:

$$\begin{aligned} \max_{w_n} & \left(w_n s_n(\mathbf{p}) \mathcal{S} - \sum_{l \in \{\mathcal{L}_m \setminus Z_j\}} w_l \frac{s_l(\mathbf{p}) \frac{\mathcal{S}}{\Theta^m}}{1 - s_n(\mathbf{p}) \frac{\mathcal{S}}{\Theta^m}} s_n(\mathbf{p}) \mathcal{S} \right)^{1-\lambda_m} \times \\ & \left((p_n - w_n - c_n) s_n(\mathbf{p}) \mathcal{S} - \sum_{h \in \{Z_j \setminus n\}} (p_h - w_h - c_h) \frac{s_h(\mathbf{p})}{1 - s_n(\mathbf{p})} s_n(\mathbf{p}) \mathcal{S} \right)^{\lambda_m} \end{aligned} \quad (2)$$

We make two assumptions for tractability. First, we assume that, for a given bilateral negotiation, all other negotiations are held fixed, commonly referred to as “Nash-in-Nash” and developed by Horn and Wolinsky (1988). Second, we assume that firms view downstream prices as fixed when bargaining over wages. Note that in equilibrium, product prices $\mathbf{p}$ will depend on negotiated wages, but this relationship is not explicitly modeled in the bargaining game due to tractability (see e.g. Draganska et al. (2010)). This means that product prices and wages are set simultaneously in equilibrium.

Note that unlike Horn and Wolinsky (1988), which assumes worker disagreement payoffs are exogenously given, in our model, the disagreement payoffs are a function of wages that are also the outcome of simultaneous bargaining.

Taking the derivative with respect to w_n and rearranging yields:

$$\lambda_m \left(w_n - \sum_{l \in \{\mathcal{L}_m \setminus Z_j\}} w_l \frac{s_l(\mathbf{p}) \frac{\mathcal{S}}{\mathcal{S}^m}}{1 - s_n(\mathbf{p}) \frac{\mathcal{S}}{\mathcal{S}^m}} \right) = \quad (3)$$

$$(1 - \lambda_m) \left((p_n - w_n - c_n) - \sum_{h \in \{Z_j \setminus n\}} (p_h - w_h - c_h) \frac{s_h(\mathbf{p})}{1 - s_n(\mathbf{p})} \right)$$

A non-linear solver may be used to solve the system of $2|N|$ equations characterized by equations 1 and 3 for w_n^*, p_n^* . Post-merger equilibrium prices can be solved for by changing the ownership structure, or the set of products included in each set Z_j .¹⁵

4 Model Implementation

In this section of the paper, we show how our model may be used to forecast the labor and product effects of mergers in different labor and product market configurations that frequently confront antitrust agencies. For example, when firms produce non-tradable goods,

¹⁵Although we have written the bargaining game as taking place between a union and a firm, under the Leontief production and worker homogeneity assumptions the model is equivalent to a bargaining game between a firm and a worker.

such as healthcare or retail services, the geographic markets for labor and products often coincide. Alternatively, many mergers involve tradable goods producers, such as meat packers or manufacturers, where the geographic market for final goods may be larger than the labor markets where the products are produced. In all cases, our model predicts that firms earn rents in the product market, and wage heterogeneity is driven by workers sharing in these rents (Abowd et al., 1999). This is a complementary approach to the classical monopsony literature in which wages and output are driven by labor supply, and allows us to predict how wages and employment are affected by strategic interaction in the product market.

In our model, the impact of mergers on workers and consumers varies depending on the configuration of the geographic markets for labor and the products sold by the merging firms. To illustrate our model’s predictions, we use a simulation approach where we calibrate markets using real world data corresponding to three configurations of the geographic market for products and labor. As we show, our model can predict the labor market effects of mergers using information commonly available to antitrust economists working on merger investigations: market shares, product prices and costs, and wages. In general, this type of detailed firm and worker level data is nonpublic and unavailable for research. We have, however, identified two public use data sets that contain sufficient information to illustrate how our model may be used to simulate merger effects for tradable and non-tradable goods. We use data from the U.S. hospital industry to simulate non-tradeable good mergers that either directly affect both the product and labor market or only the labor market. We then use data from the Colombian Manufacturing Census to examine a subset of possible tradeable good mergers that affect the product market but do not change workers’ local employment choices.

These simulations are, by necessity, somewhat stylized. For example, as we describe below in more detail, we use designations from the Dartmouth Atlas of the areas hospitals draw their patients from as proxies for the geographic markets for the labor (nursing and pharmacy

workers) and products (inpatient services) affected by hospital mergers.¹⁶ We make this choice because it provides us with a simple principled method to generate a large number of geographic markets for labor and products required for merger simulation. While we think these candidate markets reasonably approximate the competitive environment facing firms and workers, in some cases the actual antitrust markets corresponding to a hypothetical merger may be larger or smaller than those we specify. In an actual merger investigation, antitrust economists undertake a detailed market definition analysis like that described in the Merger Guidelines to more formally delineate the contours of the geographic markets for labor and products.¹⁷ As a result, the results of our simulations should be viewed as an approximation of the competitive impact of mergers on workers and consumers.

In the remainder of this section, we describe how we construct the geographic and product markets for the different market configurations examined in the simulations. Next, we describe how we use data from the U.S. hospital industry and Colombian manufacturing industries to calibrate the model. Finally, we present the findings from our simulations.

4.1 Market Configurations

The first step in simulating the price and wage effects of mergers is to specify the geographic and product markets for both the good being produced and the type of labor used in production. In our simulations, we focus on three configurations that illustrate the different mechanisms whereby mergers can affect negotiated wages in our model. The first corresponds to the non-tradable goods case when the geographic market corresponding to the product market and the labor market completely overlap (see Panel 1 of Figure 1). That

¹⁶The Dartmouth Atlas uses claims data from Medicare to identify the service areas (the regions hospitals draw patients from) of different types of hospitals. They have developed a methodology to identify local health care markets (health service areas (HSA) for standard inpatient care) and regional health care markets (hospital referral regions (HRR) for more complicated inpatient care). See here for more methodological details describing how these markets are defined. <https://www.dartmouthatlas.org/research-methods/>.

¹⁷For example, in a hospital merger case, antitrust economists frequently use patient discharge data and a method like that described in Raval et al. (2017) to identify the geographic markets for inpatient services. For a detailed discussion of market definition, see the Merger Guidelines available at: https://www.ftc.gov/system/files/ftc_gov/pdf/2023_merger_guidelines_final_12.18.2023.pdf.

is, the workers producing the product live and work in the same region where the product is consumed. This case likely corresponds to many non-tradable goods industries, such as the inpatient hospital market we examine, where the final good (health services) is produced and consumed in the same region. In this scenario, the merger of firms 1 and 2 will affect both the product market and the labor market. We define the product market as hospitals providing acute inpatient care, such as childbirth, major surgeries, or the treatment of serious illnesses.¹⁸ In general, patients have strong preferences to receive hospital services relatively close to where they live, e.g., Gowrisankaran et al. (2015). For this reason, we define the geographic market for inpatient hospital services to be the Hospital Service Area (HSA) which is defined as the collection of zip codes whose residents receive most of their hospitalizations from the hospitals in that area.¹⁹

¹⁸Antitrust agencies often define a product market to be inpatient general acute care services provided by hospitals, see, for example the complaint from a recent FTC merger challenge (Novant and Community Health Systems), see, especially, page 4.

¹⁹See <https://www.dartmouthatlas.org/faq/>.

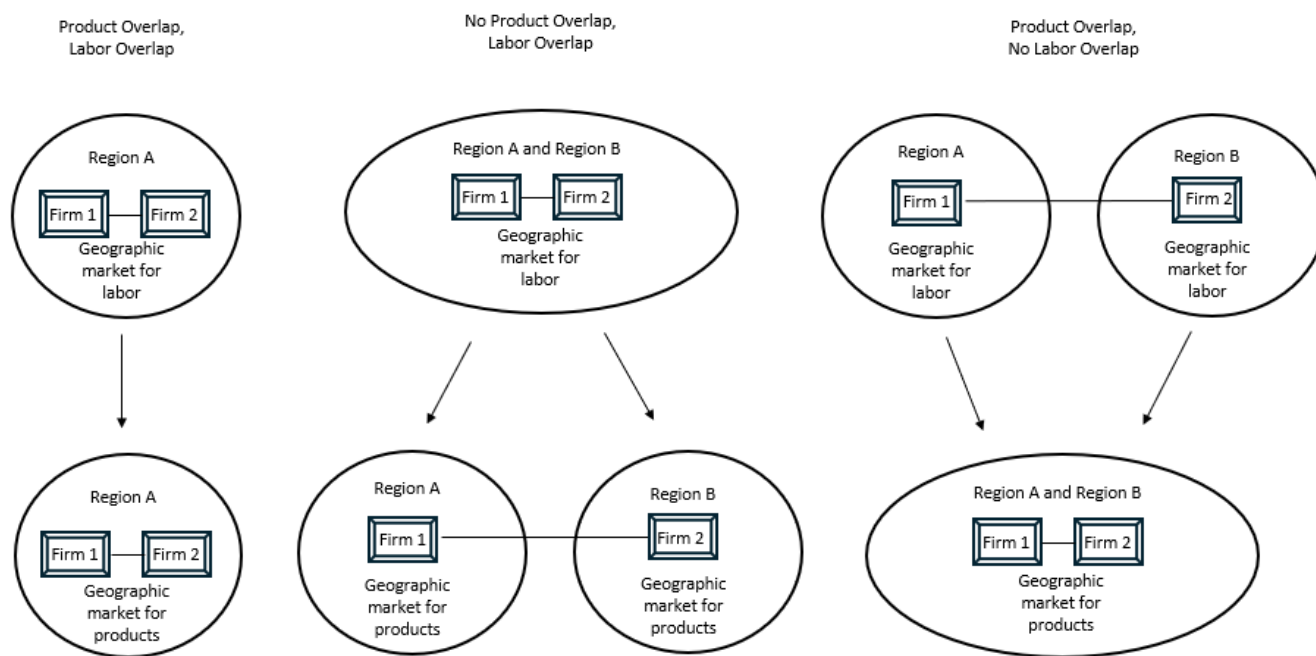


Figure 1 Product and labor market configurations.

We next specify the type of labor affected by the merger and the size of the geographic market for that type of labor. While hospitals directly employ many different types of employees, it is unlikely that all workers would be equally affected by mergers. In particular, workers with human capital specific to the hospital industry would likely benefit more from competition between potential hospital employers than unskilled workers (e.g. janitors) that could provide similar services in other industries. Prager and Schmitt (2021b) found that hospital mergers in concentrated markets caused the wages of highly skilled workers, especially those with human capital specific to medical industries (nursing and pharmacy employees), to fall the most, while the wages of unskilled workers with general human capital were unaffected. For this reason, we define the labor market affected by the merger to that for nursing and pharmacy employees. In the first scenario, we define the geographic size of the labor market to be identical to that for the product market, the HSA, and only consider the mergers of hospitals located in the same HSA. This assumption implies that workers are willing to travel as far for employment as patients are willing to travel for inpatient hospital care.

The second non-tradable goods case we examine is mergers of firms that do not compete in the same product market but do compete in the same labor market (Panel 2 of Figure 1). In this case, the merger can affect the wages of workers, but only through changes in the labor market, as there is no reduction in product market competition because firms are selling to consumers in distinct geographic markets.²⁰ We again use data from the hospital industry to simulate mergers of this type. As in the first case, the product market is defined to be inpatient hospital services where the geographic market is defined as the HSA. We also assume that the labor market that is potentially affected by the merger is that for nursing and pharmacy employees. The distinction between the first and second case is that we now assume the geographic market for labor is larger. Specifically, we assume workers consider

²⁰Note that for illustrative purposes we assume firms do not compete at all in the product market, but the model could accommodate scenarios where there is some product market overlap but more labor market overlap.

all hospitals within a Hospital Referral Region (HRR) as being potential employers. The HRR is an aggregate of contiguous HSAs and often corresponds to a metropolitan area.²¹ In this scenario, we consider mergers of hospitals in different HSAs (different product markets) that are located in the HRR (the same labor market).

The final case we examine are mergers in tradable goods where there is no overlap in the labor market, but there is an overlap in the final goods market (Panel 3 of Figure 1). Here, we limit the merger simulations to those involving merging firms operating in the same industry but that hire labor in different regions and do not compete directly for workers.²² In this case, the merger can affect the wages of workers, but only through changes in the product market (less output, but higher profits). This case most clearly corresponds to tradeable goods such as manufactured products. We analyze this case using data from the 1991 Colombian Manufacturing Census.²³ This data set identifies the industry (as measured by an SIC code) of approximately 7,000 plants located in 27 different regions of Colombia. In simulating mergers for this scenario, we define the product market as the 4-digit SIC code assigned to a plant and assume that the geographic market for products is Colombia.²⁴ We define the labor market to be skilled workers in the industry the merger takes place in.²⁵ Further, we assume that workers do not consider employers outside of their region when negotiating wages. That is, the geographic market for labor is defined to be the region the plant is located in. An antitrust economist could find narrower or broader markets, but a

²¹HRRs correspond to the region that serves patient needing highly specialized care, such as major cardiovascular surgical procedures and neurosurgery, that can only be provided by large hospitals with specialized departments. Typically, patients requiring standard care will choose a hospital within their HSA and those needing specialized care will consider the subset of hospitals capable of providing that care within the HRR. The Dartmouth Atlas defines 3436 HSAs in the U.S. (most with a single hospital) and then aggregates those HSAs into 306 HRR corresponding to the larger regions that correspond to the geographic region for specialized care.

²²Our model can also accommodate the case where firms hiring labor in the same labor market sell into a common tradeable goods market. However, to illustrate how mergers only affecting the product market can affect workers, we focus on the case of only product market overlap for tradeable goods producers.

²³This data has previously been used in studying trade, e.g., Fernandes (2007), and estimating markups, e.g., Raval (2023).

²⁴We exclude exports from the calibration.

²⁵This assumption is consistent with Prager and Schmitt (2021b) who find that skilled workers are most likely to be adversely affected by mergers.

regional labor market might not be an unreasonable candidate market, especially where data may not be forthcoming. The mergers we consider are of plants in the same industry but different regions within Colombia.

4.2 Parameter Calibration

Having specified the markets for products and labor, we now show how we can use observable information describing markets to construct the variables to calibrate the model’s parameters. Our first task is to identify the demand parameters α and δ_j as a function of margins and shares, assuming the market is in the pre-merger equilibrium described by our model. To do so we exploit Equation 1, the employer’s first order condition. With at least one employer’s margin and market share, this equation allows us to recover the demand parameter reflecting customer sensitivity to price, α .²⁶ Given α , we can then recover equilibrium margins for all other firms in the market using observed market shares for each firm and each firm’s first order conditions. Then, given observed cost information and model predicted margins, we infer product prices. Demand parameters δ_j reflecting the relative customer demand for each product are inverted from market shares, prices, and α .

Next, we turn to the bargaining first order condition in Equation 3 above to infer λ . Employer gains from trade can be calculated from employer equilibrium margins. We assume that workers are equally productive at any firm within the market, with wages varying based on sharing rents, but receive a lower wage if they take a job outside of the market. Thus worker gains from trade are given by the difference between observed wages, and outside-of-market wages, which we impute to be the mean wage in the market multiplied by a discount factor that is setting-specific, and described in Appendix A. Because our model does not include any non-wage preferences over jobs, this calibration may overstate λ , since in-market and out-of-market wages may be very similar. We allow the bargaining parameter λ to vary by product market based on the ratio of the firms’ and the workers’ gains from

²⁶We take the margin of the largest firm for which margin is available.

trade.

4.3 Data Inputs

We obtain our inputs from two primary data sources. The first is the Centers for Medicare and Medicaid Services’ (CMS) Healthcare Cost Report Information Center (HCRIS).²⁷ Under federal law, all hospitals that bill Medicare are required to provide HCRIS with annual reports including major types of costs, revenues, number of patient admissions, and numbers of different types of employees. Thus, this data source allows us to identify virtually all acute care inpatient hospitals in the U.S. The major data constraint we face is having sufficient information describing hospitals to calibrate the model as described above. In our application, this requires that we observe firm level market shares²⁸ and wages for each significant hospital, and the margin of at least one hospital, in a specified geographic market. In our simulations, we focus on the set of hospitals observed in 2006 as in this year we have complete data for a large fraction of hospitals. We also use information from the American Hospital Association’s annual survey to identify the owner of a hospital (e.g. if the hospital is in a system) and the HSA the hospital is located in.

Our second primary data source is the 1991 Colombian Manufacturing Census provided by the Departamento Administrativo Nacional de Estadística between 1981 and 1991. The data source contains plant level information including the number of employees of different types (e.g., unskilled, skilled, and managerial),²⁹ total compensation paid to each type of worker, measures of fixed and variable costs, total domestic sales, and total sales for exports.

²⁷We use a data set that was cleaned and used by Prager and Schmitt (2021b) and is available at <https://www.openicpsr.org/openicpsr/project/120834/version/V1/view> (Prager and Schmitt, 2021a).

²⁸We define inside labor shares according to the relative employment of nursing and pharmacy workers across firms as reported in HCRIS. We take the number of nursing and pharmacy workers, defined as SOC codes 29-1051, 29-1141, 29-1171, and 29-2061, employed outside the hospital industry, defined as NAICS code 622000, in the BLS occupation and employment statistics data, to calculate a worker outside option share across all markets of 52%.

²⁹We measure the share of labor in a market as the fraction of skilled labor at each plant in a given industry (4-digit SIC code) within the Department, the narrowest geographic region reported in the data source, corresponding to 27 regions of Colombia. We set the worker’s outside share to 52% to allow consistent comparison of worker welfare to the hospital data.

Table 1: Pre-Merger Summary Statistics for Simulated 2006 Health Service Area Mergers (n=855), Products (n=1044), Markets (n=325), Product and Labor Market Overlap

	50%	5%	25%	75%	95%
Calibration Inputs					
Margin (%)	58	25	47	68	78
Hospital Cost per Worker (\$100k/year)	30	14	23	41	94
Number of Workers per Hospital	30	1.3	8.8	63	162
Hospital Wage (\$100k/year)	0.65	0.51	0.59	0.71	0.81
Calibration Outputs					
Hospital Outside Wage (\$100k/year)	0.39	0.32	0.36	0.41	0.46
Hospital Margin per Worker (\$100k/year)	28	5.1	14	52	116
Hospital Price per Worker (\$100k/year)	65	28	46	96	187
Market Price Sensitivity (α)	-0.069	-0.39	-0.14	-0.037	-0.014
Merger Conditions					
Market Bargaining Power (λ)	0.99	0.97	0.99	0.99	0.99
Merger HHI_L and HHI_P	4,371	2,093	2,917	5,716	8,821
Merger ΔHHI_L	798	131	284	2,379	4,781
Merger ΔHHI_P	798	131	284	2,379	4,781

In addition, the data source identifies the primary industry the plant is in, and the region of Colombia that the plant is located in. Unfortunately, the data does not identify the owner of a plant. This means we cannot observe if multiple plants have a common owner. Consequently, we assume that pre-merger, each plant is a sole proprietorship.

Further details regarding how we constructed employee wages as well as firm-level margins for both primary sources may be found in Appendix A

4.4 Calibration Inputs and Outputs

Tables 1-3 contain descriptive statistics describing the calibration inputs and outputs used in our simulations. Table 1 describes the first scenario where we analyze hospital mergers that have both a product and geographic overlap. Our sample consists of 325 HSA markets with a total of 1044 hospitals (products).

To construct the simulation sample, we examine all pairwise combinations of hospital systems and hospitals likely to have a meaningful impact on competition. Specifically, we

restrict our analysis to mergers where the pre-merger HHI is at least 1500 and the merger increases the HHI by at least 100. This selection criterion, which generally aligns with the thresholds for structural presumptions used in agency guidelines, yields a final sample of 855 simulated mergers.³⁰

We highlight a few important characteristics from the table. First, hospitals tend to have relatively high margins: in the median market margins are calculated to be 58% using the approach from Dafny (2009) and Garmon (2017). Second, the mergers we simulate lead to large increases in market concentration in already concentrated markets. The median merger increases $HHI_P (=HHI_L)$ by about 798 from a pre-merger HHI_P (HHI_L) of 4371. Third, our model predicts that firms have most of the bargaining power: λ is estimated to be very close to 1. This low level of worker bargaining power, however, does not imply that workers do not gain from bargaining. Because the margin the typical hospital earns per nursing or pharmacy worker is large, roughly \$2.8 million per worker, capturing even a small amount of surplus can dramatically benefit nursing and pharmacy workers. The imputed out-of-market wage for a nursing or pharmacy worker at the median hospital is \$39,000, compared to the hospital wage of \$65,000, meaning that workers are earning over 50% more through the bargaining process relative to a take-it-or-leave-it offer, even with λ approaching 1.³¹

Table 2 describes the second scenario where we examine mergers that affect the labor market but not the product market. That is, where hospitals in the same HRR but in different HSAs merge. In this case, we simulate the impact of 324 mergers taking place in 62 HRRs.³² The calibration inputs and outputs in this scenario are very similar to those above because we are using data from the same industry. The primary difference is that the

³⁰See Appendix A for further details on the set of mergers.

³¹While nursing and pharmacy workers are an important input into inpatient care, they account for a relatively low share of a hospital’s marginal costs, roughly 2%. At the median hospital, a nursing or pharmacy worker is estimated to earn \$65,000 a year while the hospital’s other marginal costs per nursing or pharmacy worker are estimated to be about \$3 million.

³²The scenarios we have specified are consistent with the geographic configurations of mergers that have taken place. Cooper et al. (2019) contains a census of hospital mergers taking place in the U.S. between 2001 and 2014. Of the 605 hospital mergers Cooper et al. have identified and for which we have geographic information, 101 mergers involved hospitals operating in the same HSA, and 331 involved hospitals located in the different HSAs but in the same HRR.

Table 2: Pre-Merger Summary Statistics for Simulated 2006 Hospital Referral Region Mergers (n=324), Products (n=526), Markets (n=62), No Product Market Overlap, Labor Market Overlap

	50%	5%	25%	75%	95%
Calibration Inputs					
Margin (%)	60	26	49	68	81
Hospital Cost per Worker (\$100k/year)	30	14	22	40	92
Number of Workers per Hospital	26	1.2	8.1	59	178
Hospital Wage (\$100k/year)	0.64	0.52	0.59	0.71	0.82
Calibration Outputs					
Hospital Outside Wage (\$100k/year)	0.38	0.34	0.36	0.4	0.46
Hospital Margin per Worker (\$100k/year)	25	5.1	13	48	100
Hospital Price per Worker (\$100k/year)	61	27	46	90	179
Market Price Sensitivity (α)	-0.081	-0.4	-0.15	-0.039	-0.015
Merger Conditions					
Market Bargaining Power (λ)	0.99	0.98	0.99	0.99	0.99
Merger HHI_L	2,428	1,579	2,163	3,082	4,673
Merger ΔHHI_L	357	110	171	691	1,869
Merger ΔHHI_P	0	0	0	0	0

mergers have a much smaller impact on the labor market. Because HRRs have many more hospitals in them than HSAs, both the level of market concentration and change in market concentration are much smaller for these hypothetical mergers. The median merger raises HHI_L by only 357 and the median pre-merger HHI_L is only 2428.³³

Table 3 describes the calibrated inputs and outputs from the Colombian manufacturing mergers which, because they are cross-region mergers, directly affect the product market but not the labor market. There are many differences between the manufacturing industries in the Colombian data and the U.S. hospital data. First, the manufacturing data consists of firms operating in many (53) different industries. While there is wide variation in the margins firms earn in the different industries, virtually all of them are considerably lower (median of 26%) than U.S. hospitals. Moreover, skilled workers appear to have more relative bargaining power in these industries than the U.S. hospital industry (median $\lambda=.92$). Skilled

³³The reported concentration here refers only to the labor market and not the product market which is not directly affected by the merger.

Table 3: Pre-Merger Summary Statistics for Simulated 1991 Colombia Manufacturing Mergers (n=423), Products (n=252), Markets (n=41), Product Market Overlap, No Labor Market Overlap

	50%	5%	25%	75%	95%
Calibration Inputs					
Margin (%)	26	7.5	18	36	49
Plant Cost per Worker (\$100k/year)	0.67	0.14	0.37	1.3	4.2
Number of Workers per Plant	70	7	32	140	362
Plant Wage (\$100k/year)	0.063	0.03	0.047	0.081	0.11
Calibration Outputs					
Plant Outside Wage (\$100k/year)	0.029	0.006	0.02	0.035	0.043
Plant Margin per Worker (\$100k/year)	0.26	0.077	0.14	0.53	0.99
Plant Price per Worker (\$100k/year)	1.2	0.35	0.75	1.9	4.5
Market Price Sensitivity (α)	-5.1	-15	-8.1	-2.5	-1.2
Merger Conditions					
Market Bargaining Power (λ)	0.92	0.71	0.88	0.96	0.97
Merger HHI_P	1,955	1,559	1,708	2,629	3,794
Merger ΔHHI_L	0	0	0	0	0
Merger ΔHHI_P	334	115	196	582	1,402

labor also accounts for a considerably higher fraction of firm’s marginal costs, around 13%. Finally, the mergers we simulate tend to take place in less concentrated product markets than in the U.S. hospital industry, with a median pre-merger HHI_P of roughly 2000.³⁴

4.5 Simulation Results

To illustrate how the different mechanisms in our model affect equilibrium outcomes following mergers, we simulate mergers corresponding to three different wage and price setting models. The first model assumes that wages are exogenously determined and that mergers only affect competition in the product market; that is, there is no bargaining over wages. We refer to this as the “Downstream Only” model that corresponds to the traditional Bertrand merger simulation model that is the workhorse of modern horizontal merger analysis, see, e.g., Werden and Froeb (1994). The next case we consider is a bargaining model where the threat

³⁴Again, the reported HHI_P corresponds to the product market, which experiences a change in market structure under this market configuration, unlike the labor market.

Comparison of Welfare Effects Across Different Model Configurations and Specifications

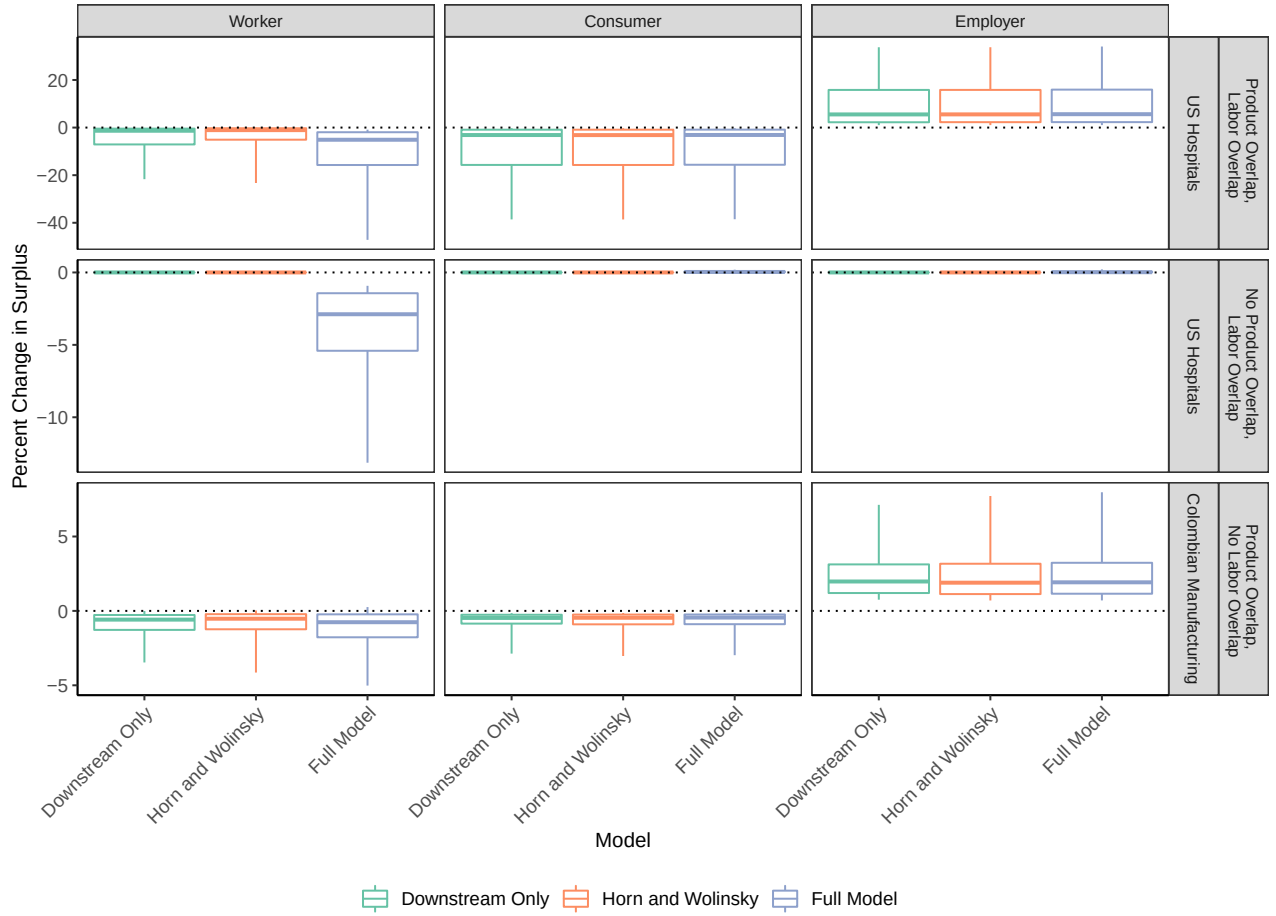


Figure 2 displays worker, consumer, and employer surplus effects of the mergers in percentage terms, including the effect of changes to shares. Each box and whisker plot shows the 5th, 25th, 50th, 75th, and 95th percentiles of outcomes among the simulated mergers. We show results for three models: downstream only, bargaining with fixed worker threat point (Horn and Wolinsky) and the full bargaining model.

point is exogenously determined as in Horn and Wolinsky (1988). In our implementation of Horn and Wolinsky (HW) we set the threat point for workers to be the average share-weighted wage in the local labor market (including the outside employment option) instead of zero.³⁵ Finally, we simulate mergers using our model where the wages of all firms in each labor market are determined by bargaining and where, post-merger, the workers' threat point no longer includes either merging party. We refer to this as the "Full Model."

The results of the merger simulations are shown in Figure 2 in a series of box and whisker plots. Each row of the figure corresponds to one of the three configurations of the product and labor markets for which we simulate mergers: (1) where the merger takes place in the same geographic market for labor and the final good market, (2) where the merger affects the local labor market but not the product market, and (3) where the merger only directly affects the product market. For each type of simulated merger, we show the distribution of welfare effects for worker, consumer, and employer welfare in the columns of the figure where the median is shown by a solid line, the interquartile range is shown by the upper and lower bounds of the box, and the endpoints of the whiskers correspond to the 5th and 95th percentiles.³⁶

We begin by describing the simulated effects of hospital mergers that directly affect the labor and the product market. The downstream only model is a good proxy for what an antitrust agency might predict if faced with making a decision on a merger based on the product market alone. Since wages are fixed, the only potential impact on workers in this model is through a reduction in output (and labor employed). In Figure 2, we see that the merger simulations corresponding to the downstream only model have very small effects on

³⁵In HW the threat point wage offer to workers was set to zero (implicitly assuming no worker surplus should negotiations fail). We think that setting the outside wage equal to the average wage in the labor market is a more realistic depiction of the outcome open to workers should negotiations fail and allows for better comparison with our full model.

³⁶Change in consumer surplus is defined as $100 * \frac{\frac{\log(\sum_j e^{\delta_j + \alpha p_j'})}{\alpha} - \frac{\log(\sum_j e^{\delta_j + \alpha p_j})}{\alpha}}{\frac{\log(\sum_j e^{\delta_j + \alpha p_j})}{\alpha}}$. Percent change in employer surplus is defined as $100 * \frac{\sum_j s_j' m_j' - \sum_j s_j m_j}{\sum_j s_j m_j}$. Percent change in worker surplus is defined as $100 * \frac{\sum_j s_j' w_j' - \sum_j s_j w_j}{\sum_j s_j w_j}$.

worker welfare. By contrast, we see that mergers substantially reduce consumer surplus and increase producer surplus due to standard product market recapture effects.

The next simulation (HW) allows for bargaining between an employer and its workers, but assumes that the outside wage is fixed (exogenously determined). This means that the merger cannot affect the threat point of the workers either positively (through increased quantity produced at non-merging firms) or negatively (through decreased wages at non-merging firms). We see little difference in predicted worker welfare between the downstream only simulations and the HW simulations. Harm to workers is, on average, a bit lower in HW, however, in extreme cases the HW model predicts more harm than the downstream only model.³⁷

In moving to the full model described in Section 2, we see notable differences in the labor market outcomes for workers. Workers are much more adversely affected by the merger when the merger is allowed to affect their threat point. In these simulations, the downstream only and HW models only explain about 26% and 23%, respectively, of the worker harm predicted by the full model.³⁸ Our model shows that if wages are determined by bargaining, mergers can have a substantial impact on workers by reducing their threat point in negotiations with employers.

At first glance, it might seem surprising that the simulations do not show much change in either consumer or employer surplus despite the large change in worker surplus. There are two related reasons for this outcome, both specific to our examination of the hospital industry. First, the nursing and pharmacy employee wages, in aggregate, make up a very small share of total share of hospital costs, roughly 2%. As a result, the transfer from workers to the employer post-merger has a small incremental impact on hospital profits relative to the

³⁷When comparing the HW model to the downstream only model: 1) employer bargaining power increases 2) employer profits increase the total surplus in the bargaining game. The HW model therefore predicts higher or lower wages when wages are allowed to adjust post-merger.

³⁸For each simulation, we have calculated the ratio of the change in worker surplus from the downstream only and HW model relative to the full model. We find that for the median merger the ratio of the reduction in worker welfare from the downstream only and HW model relative to the full model is 0.26 and 0.23, respectively.

transfer from consumers to producers. Second, as shown in Table 1, workers have extremely limited bargaining power. As a result, pre-merger most of the surplus from the gains from trade between workers and firms is accrued by firms. However, as shown in Appendix C Figure 4, in simulated markets where worker bargaining power is much higher and the labor share of costs is larger, we show that the pass-through of lower workers' wages attenuates the effect of mergers on consumers. In fact, when mergers have only an overlap in the labor market, mergers increase consumer surplus.

In the second row of Figure 2 we examine the changes in worker, consumer, and employer surplus resulting from hospital mergers that directly affect the labor market but not the product market. These are mergers of hospitals that are assumed to not compete in the product market (located in different HSAs) but do compete in the labor market for nursing and pharmacy workers (hiring workers from throughout the HRR). In this scenario, by construction, the downstream only and HW model predict that there should be no change in wages, and as a result no change in producer or consumer welfare. In the full model, however, we see that the reduction in an independent employment option for workers does reduce worker welfare, albeit by considerably less than the first case.³⁹

Finally, we examine the third case where the merger directly affects the product market but not the labor market. In this scenario, we simulate the impact of mergers of Colombian manufacturing plants in the same industry (4 digit SIC code), but located in different regions of Colombia. In these simulations, we assume that firms are competing in the same national final goods market, but are hiring workers at plants located in different regions. As was the case in first scenario, we see predicted reductions in worker welfare resulting from a reduction in labor demand from the merger. However, the predicted effects of the three models are more similar than in the other market configurations, in which there was labor overlap. For instance, for the median merger the downstream only model and HW model account for 68%

³⁹The mergers taking place here (HRR markets) are by construction less concentrated than those in the first example. As a result, the average effect of mergers on workers in this sample of simulated mergers is smaller.

and 64% of the reduction in worker welfare predicted by the full model.^{40, 41} This finding suggests it is primarily the loss of an employment option in the worker’s local labor market that causes our full model to predict greater consumer harm than the standard downstream only or HW models.

Overall, for the set of manufacturing mergers we examine the reduction in worker welfare is fairly small: the median merger reduces worker welfare by less than 1% in all three of the models. This result is in part due to the fact that these mergers result in smaller changes in market concentration than in the simulated hospital merger, and in part because of the limited effect of product market overlap on the bargaining outcome. We further disentangle these mechanisms below. As was the case with the simulated hospital mergers, because workers are estimated to have relatively low bargaining power and total labor costs are, on average, only 13% of Colombian manufacturers’ costs, the pass through of reduced wages into either consumer or producer welfare is quite small.

The findings shown for producer and consumer surplus in Figure 2 are consistent with traditional models of merger simulation that focus on product market competition. As has been shown elsewhere, e.g. Nocke and Whinston (2022), absent efficiencies, mergers that result in significant increases in product market concentration (ΔHHI_P) reduce consumer surplus and increase employer surplus. However, to date, there is little model based evidence relating the economic significance of mergers as measured by the change in either product or labor market concentration (ΔHHI_P or ΔHHI_L) to changes in worker wages, consumer prices, and employment. We now examine how those changes manifest themselves in terms of changes in market wages, employment (output), and consumer prices, which are

⁴⁰For each simulation, we calculated the ratio of the change in worker surplus from the downstream only and HW model relative to the full model. We find that for the median merger the ratio of the reduction in worker welfare from the downstream only and HW model relative to the full model is 0.68 and 0.64, respectively.

⁴¹When there is only product market overlap, the only change between the HW and full model is the endogeneity of the worker threat point. That is, in equilibrium workers’ threat point changes. In this scenario, whether the change in the threat point increases or decreases worker surplus depends on market characteristics.

other common measures of merger impact.⁴² In some scenarios, our model predictions are ambiguous for workers, as wages can go up or down, depending on the bargaining mechanisms discussed above. Worker surplus, defined as the total wage bill, therefore may mask heterogeneity—while some workers will lose their jobs, wages may go up for those who remain in the labor market.

In Figure 3 we begin by examining hospital mergers affecting both the product and the labor market that resulted in the largest reductions in worker surplus. In this case, a merger results in three changes that can harm workers: lower wages due to a reduction in worker leverage (loss of an employment option), lower wages due to an increase in firm leverage (the firm is able to make sales at another hospital in the event negotiations breakdown), and a reduction in employment resulting from increased hospital prices (recapture). Workers could benefit from the merger, however, if they were able to capture some of the increased employer profits caused by the reduction in product market competition. In the first row of Figure 3, we see that all simulated mergers result in a reduction in both worker wages and employment (reduced output), and that reduction increases as the impact on competition (ΔHHI_P or ΔHHI_L) increases. For hospital mergers that only affect the labor market but not the product market, the merger only affects workers by reducing their leverage. In our simulations, this manifests itself as a reduction in market wages leading to a very small increase in employment (row 2 of Figure 3). Were wages to make up a larger share of a hospital’s costs, pass-through to prices would be higher resulting in a more tangible increase in employment.⁴³

The third case, where the merger increases concentration in the product market but not the labor market, is perhaps the most interesting. Here, mergers result in all the same impacts as the first case (row 1 of Figure 3), with the exception that there is no reduction in worker leverage, due to the lack of labor market overlap. The mergers still harm workers by

⁴²In our model, by assumption, output is proportional to employment.

⁴³In Appendix C Figure 4 we conduct additional simulations where we assume workers’ have more bargaining power and the worker share of costs, margins, and prices are adjusted accordingly.

Relationship Between Change in Wage, Price, and Output and ΔHHI

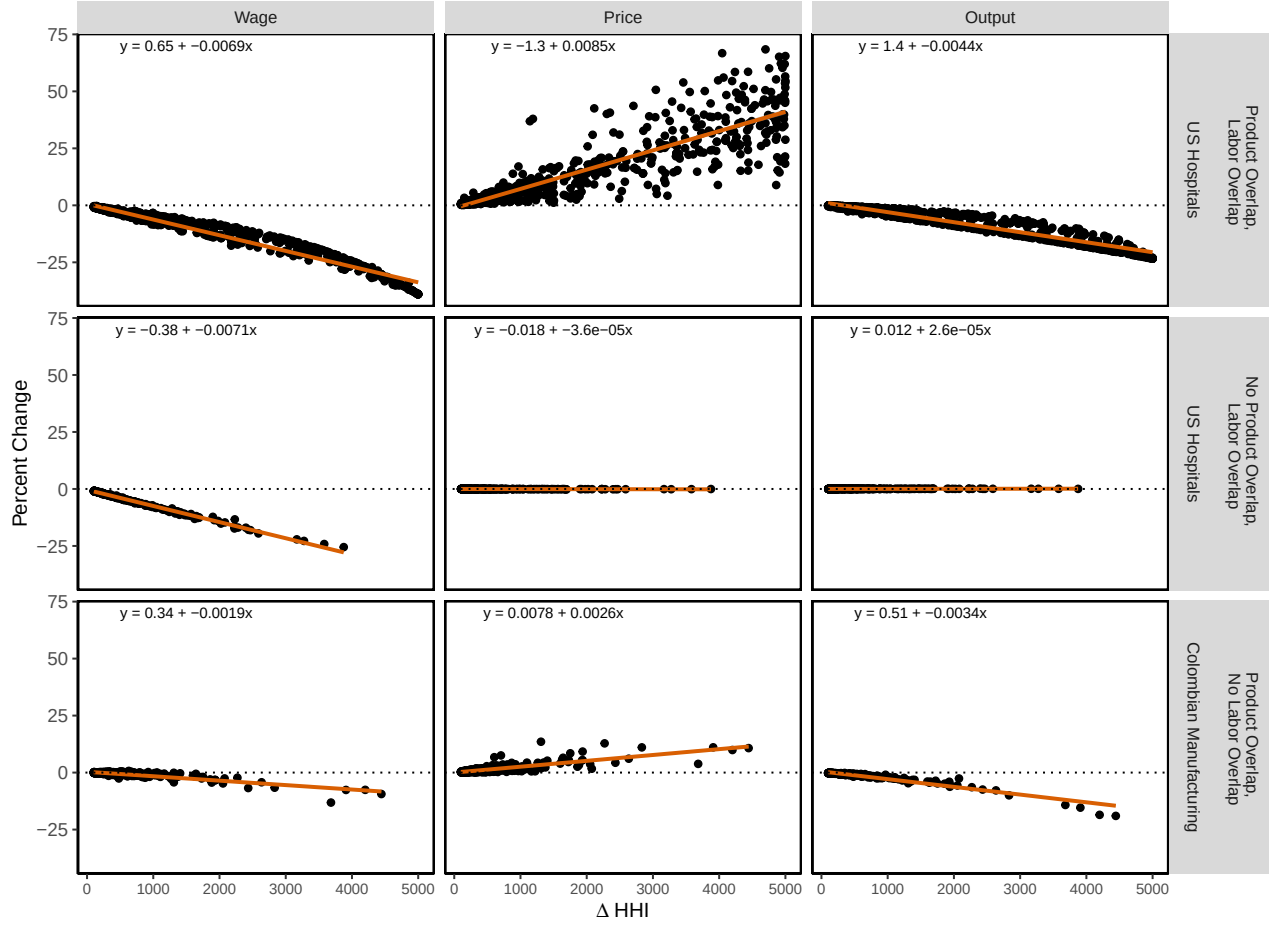


Figure 3 displays wage, price, and output effects of the mergers in percentage terms, weighting by pre-merger shares. Each dot represents one merger. All effects are plotted against ΔHHI , where ΔHHI in the first row is ΔHHI_P which is equal to ΔHHI_L , in the second row ΔHHI_L since $\Delta\text{HHI}_P = 0$, and the third row ΔHHI_P since $\Delta\text{HHI}_L = 0$.

lowering output through a traditional unilateral effect and increasing the firm's leverage in wage negotiations, but the merger also results in a price increase that increases the rents to be split between workers and the firm that may increase wages. In the third row of Figure 3, we see that for mergers with relatively small increases in concentration ($\Delta\text{HHI}_P < 1000$), the wage effects of mergers are very small and in some cases actually positive. This suggests that for small changes in ΔHHI_P the wage bargaining effects of the merger effectively offset each other. However, as mergers become more anticompetitive ($\Delta\text{HHI}_P > 1000$), simulated mergers almost always lower wages in addition to employment.

5 Screens

The results from the merger simulations show that if a bargaining model like ours describes wage setting, then mergers that affect the relative bargaining position of workers and their employers can have large impacts on worker welfare. As a result, merger simulations generated using models focusing only on downstream competition may dramatically underestimate the impact of consummated mergers on worker welfare. However, while downstream only models may fail to accurately measure wage effects, it is possible that these models can still play an important role in identifying mergers that harm workers. Here, we test the efficacy of a commonly used downstream only model in screening mergers that harm workers under the three market configurations we have studied.

The results of this analysis are shown in Table 4. The first row of the table evaluates the efficacy of screens where the geographic market for labor and products are the same (HSA). In our simulations, we find that 78% of mergers that were not flagged as harmful to consumers using a downstream only model were predicted to reduce worker welfare by at least 1% using the full model.⁴⁴ However, under the same screen, only 2% of these mergers would reduce worker welfare by more than 5%. We also find that 77% of mergers that reduce worker surplus by 1% and essentially all of mergers that reduce worker surplus by 5% would be flagged as anti-competitive by the downstream-only model. Collectively, we take this as suggestive evidence that when the product and labor markets completely overlap, as is often the case for non-tradeable goods, that product market screens likely identify the mergers most harmful to workers.

Similarly, we find that product market screens are very effective in identifying mergers that harm workers where there is only a direct overlap in the product market, as is common in tradeable goods markets. Finally, if there is no product market overlap, by construction, a downstream only merger screen would never identify a merger as harmful to workers. Hence,

⁴⁴In our screen, we flag all mergers that are predicted to lower consumer surplus by at least 1% as anticompetitive.

Table 4: Fraction of mergers that harm workers missed and caught, using a downstream only model and a 1%, 5% enforcement threshold, by market configuration

Setting	Type of Merger	Mergers Missed		Mergers Caught	
		> 1%	> 5%	> 1%	> 5%
Hospitals	Product Overlap, Labor Overlap	0.78	0.02	0.77	0.99
Hospitals	No Product Overlap, Labor Overlap	0.90	0.27	0.00	0.00
Manufact.	Product Overlap, No Labor Overlap	0.28	0.00	0.45	1.00

Note:

‘Mergers Missed’ denotes the fraction of simulations where the full model predicts that a merger harms workers by either at least 1% or 5%, conditional on a downstream only model predicting no more than 1% consumer harm. ‘Mergers Caught’ denotes the fraction of simulations where the partial model predicts that a merger yields more than 1% consumer harm, conditional on the full model predicting that a merger harms workers by either at least 1% or 5%.

if there are mergers that involve firms that are important employers for a well defined type of labor but not competitors in the product market, then antitrust economists will be forced to use information about substitution patterns in the labor market rather than the product market to forecast the labor effects of mergers. We think the case we have examined here, a broader labor market than product market for hospital services, is a plausible instance where this could occur. Another potential example, might be merger of non-competing retailers who use similar employees for certain kinds of tasks. It is in these instances that antitrust economists must focus very carefully on the scope of the labor market to determine the likelihood of harm to workers as product market screens cannot provide helpful information.

6 Conclusion

We model the interaction between product and labor markets using a two-level supply chain, where employers and workers negotiate wages in a Nash-in-Nash game upstream, and firms produce differentiated products and engage in Bertrand price competition downstream. In our modeling approach, mergers can affect worker welfare through four channels: changes in workers’ or employers’ bargaining leverage, labor demanded by employers, and the rents to

be shared.

Our simulations highlight how different configurations of product and labor markets shape merger effects on workers. When mergers affect only the product market (e.g., involve manufacturing plants producing tradeable goods located in different local labor markets), mergers harm workers by increasing manufacturer’s bargaining leverage, but may also benefit workers by increasing the total rents for workers and the firm to share in negotiations. We find that these types of mergers have little effect on worker wages when the ΔHHI_P associated with the merger was less than 1000, but resulted in meaningful reductions in wages when the change in market concentration was larger. By contrast, mergers that affect the labor market reduce worker leverage in bargaining. We also found a strong negative relationship between the ΔHHI_L and wages for these types of mergers. We view this finding as illustrating how the loss of an independent employment option in a labor market can reduce workers’ wages.

Finally, we show that if wages are determined by a bargaining process where employers set wages in response to rivals in the local labor market, then downstream only models will underestimate, often substantially, the effect of mergers on workers. These product market models, however, can still play an important role in identifying the mergers likely to harm workers. We find that screening tools that focus exclusively on product market competition correctly flag the mergers that substantially harm workers when the merging employers compete in the product market. Unfortunately, by construction, these tools cannot be used when merging employers are not competitors in the product market.

In many service industries, such as retailing and healthcare, it is possible that workers have larger labor markets (commuting zones) than the geographic market(s) corresponding to the services produced by the merging firms. Consequently, a merger of firms in distinct geographic product markets may still adversely affect employees seeking employment in a common labor market. Our results suggest that for this class of mergers, antitrust authorities cannot rely on product market proxies. An interesting and potentially important question for future research is to determine if there are a significant number of mergers that materially

affect labor markets without affecting product market competition.

During the preparation of this work the authors used Google Gemini in order to edit the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

References

- Abowd, J. M., F. Kramarz, and D. N. Margolis (1999). High wage workers and high wage firms. *Econometrica* 67(2), 251–333.
- Arnold, D. (2021). Mergers and acquisitions, local labor market concentration, and worker outcomes. *Working Paper*.
- Asker, J. and V. Nocke (2021). Collusion, mergers, and related antitrust issues. In *Handbook of industrial organization*, Volume 5, pp. 177–279. Elsevier.
- Berger, D., K. Herkenhoff, and S. Mongey (2022). Labor market power. *American Economic Review* 112(4), 1147–1193.
- Berger, D. W., T. Hasenzagl, K. F. Herkenhoff, S. Mongey, and E. A. Posner (2023). Merger guidelines for the labor market. Technical report, National Bureau of Economic Research.
- Burdett, K. and D. T. Mortensen (1998). Wage differentials, employer size, and unemployment. *International Economic Review*, 257–273.
- Card, D. (2022). Who set your wage? *American Economic Review* 112(4), 1075–1090.
- Card, D., A. R. Cardoso, J. Heining, and P. Kline (2018). Firms and labor market inequality: Evidence and some theory. *Journal of Labor Economics* 36(S1), S13–S70.
- Craig, S. V., M. Grennan, and A. Swanson (2021). Mergers and marginal costs: New evidence on hospital buyer power. *The RAND Journal of Economics* 52(1), 151–178.
- Dafny, L. (2009). Estimation and identification of merger effects: An application to hospital mergers. *The Journal of Law and Economics* 52(3), 523–550.
- Draganska, M., D. Klapper, and S. B. Villas-Boas (2010). A larger slice or a larger pie? an empirical investigation of bargaining power in the distribution channel. *Marketing Science* 29(1), 57–74.
- Eslava, M., J. Haltiwanger, A. Kugler, and M. Kugler (2010). Trade liberalization and worker displacement: Evidence from trade reforms in colombia. *Mimeo*.
- Fernandes, A. M. (2007). Trade policy, trade volumes and plant-level productivity in colombian manufacturing industries. *Journal of international economics* 71(1), 52–71.
- Garmon, C. (2017). The accuracy of hospital merger screening methods. *The RAND Journal of Economics* 48(4), 1068–1102.

- Gowrisankaran, G., A. Nevo, and R. Town (2015). Mergers when prices are negotiated: Evidence from the hospital industry. *American Economic Review* 105(1), 172–203.
- Horn, H. and A. Wolinsky (1988). Bilateral monopolies and incentives for merger. *The RAND Journal of Economics*, 408–419.
- Jarosch, G., J. S. Nimczik, and I. Sorkin (2024). Granular search, market structure, and wages. *Review of Economic Studies*, 3569—3607.
- Lachowska, M., A. Mas, and S. A. Woodbury (2020). Sources of displaced workers’ long-term earnings losses. *American Economic Review* 110(10), 3231–3266.
- Manning, A. (2003). *Monopsony in motion: Imperfect competition in labor markets*. Princeton University Press.
- Marinescu, I. and H. Hovenkamp (2019). Anticompetitive mergers in labor markets. *Ind. LJ* 94, 1031.
- Naidu, S., E. A. Posner, and G. Weyl (2018). Antitrust remedies for labor market power. *Harvard law review* 132(2), 536–601.
- Nevo, A. (2001). Measuring market power in the ready-to-eat cereal industry. *Econometrica* 69(2), 307–342.
- Nocke, V. and M. D. Whinston (2022). Concentration thresholds for horizontal mergers. *American Economic Review* 112(6), 1915–1948.
- Prager, E. (2025). Antitrust enforcement in labor markets. *NBER Working Paper* (w34572).
- Prager, E. and M. Schmitt (2021a). Data and code for: Employer consolidation and wages: Evidence from hospitals. Technical report, Nashville, TN: American Economic Association [publisher], 2021. Ann Arbor, MI: Inter-university Consortium for Political and Social Research [distributor]. <https://doi.org/10.3886/E120834V1> (accessed February 3, 2023).
- Prager, E. and M. Schmitt (2021b). Employer consolidation and wages: Evidence from hospitals. *American Economic Review* 111(2), 397–427.
- Raval, D. (2023). Testing the production approach to markup estimation. *Review of Economic Studies* 90(5), 2592–2611.
- Raval, D., T. Rosenbaum, and S. A. Tenn (2017). A semiparametric discrete choice model: An application to hospital mergers. *Economic Inquiry* 55(4), 1919–1944.
- Schubert, G., A. Stansbury, and B. Taska (2021). Employer concentration and outside options. *Available at SSRN* 3599454.
- Setzler, B. (2025). Labor and product market power, endogenous quality, and the consolidation of the us hospital industry. Working Paper w34180, National Bureau of Economic Research.

- Sheu, G. and C. Taragin (2021). Simulating mergers in a vertical supply chain with bargaining. *The RAND Journal of Economics* 52(3), 596–632.
- Sokolova, A. and T. Sorensen (2021). Monopsony in labor markets: A meta-analysis. *ILR Review* 74(1), 27–55.
- Werden, G. J. and L. M. Froeb (1994). The effects of mergers in differentiated products industries: Logit demand and merger policy. *JL Econ. & Org.* 10, 407.
- Whinston, M. D. (2007). Antitrust policy toward horizontal mergers. *Handbook of industrial organization* 3, 2369–2440.

A Data Appendix

A.1 Margins

A.1.1 U.S. Hospitals

We assume that all variable inputs other than nursing and pharmacy worker wages (which are determined endogenously in our model) are purchased at constant margin cost. We calculate the margin of downstream firms using commercial price, costs, and discharges from HCRIS as cleaned by Prager and Schmitt (2021b), and case-mix index from CMS following Dafny (2009) and Garmon (2017). Margins predicted to be > 1 or $< .01$ are marked as missing. We convert the percent margin of the acquiring firm into a level margin using information on operating costs and wages from HCRIS. Note that in converting the percent margins following Dafny (2009) and Garmon (2017) calculations to level margins at the worker year level, we remove the costs associated with nursing and pharmacy worker wages from the hospital's marginal costs because those wages are endogenously determined in our model. In calibrating α , we use the margin of the largest firm in the HSA for which margin is available. From there we obtain the margins of all other firms in the market from the downstream first order conditions.

A.1.2 Colombian Manufacturing

These data report variable costs that include industrial expenditures such as parts and fuel, raw materials, wage bill and worker benefits, and total energy consumed.⁴⁵ Other expenditures such as rent and royalties we consider to be fixed. We assume that all costs are allocated between domestic and export production according to the share of exports, so, for example, the domestic wage bill for a firm that exports 10% of its sales will be 90% of the total wage bill. Total wages and benefits paid to skilled workers are excluded from our

⁴⁵We convert dollar values from Colombian pesos to US dollars using the average conversion rate of 633:1 for 1991 from FRED, available at: <https://fred.stlouisfed.org/series/COLCCUSMA02STM>.

measure of marginal costs as these are determined endogenously in the model.

We take the most recent year of data available which is 1991. We drop all industries that are non-specific, e.g. include “miscellaneous” in the title. We drop plants that have 100% of sales as exports. We also drop plants with extremely small shares ($< 3\%$), or no skilled workers, to avoid outliers. We drop plants with margins less than 1%, which occurs for 10% of the plants. We drop textiles, aircraft, and sporting goods industries because the margins of their largest firms are too low for the model to rationalize.

A.2 Wages

A.2.1 U.S. Hospitals

To calibrate our model, we need to observe wages within the market where wages are endogenously determined and also specify the outside wage. Measuring the wages of workers within the affected industry is straightforward. Using the cleaning code of Prager and Schmitt (2021b), we pull wages and employment of nursing and pharmacy employees from HCRIS for the hospital merger simulations. We define a hospital’s wage as the ratio of total wages paid for nursing and pharmacy workers assuming each full-time-equivalent works 40 hours a week in a calendar year divided by the number of full-time-equivalent nursing and pharmacy workers employed by the hospital.

Measuring the wage of the outside option for workers is more involved. Here we need to forecast the wage a worker would earn if they were forced to work outside the industry being studied. We use a different approach to forecast this outside wage to match the likely institutional characteristics of the nursing and pharmacy workers market and that of industrial workers. We assume that a worker who had to leave the hospital setting would work in a different sector but would continue to work as a nursing or pharmacy worker. However, by being forced to change sectors, the displaced worker would likely face some wage penalty when moving to the new industry. That is, the worker would not enter the new industry at that industry’s mean wage, but at some fraction reflecting some displacement effect in

the years immediately following separation. To construct this measure, we first use the BLS OES to estimate (at a national level) the relative wage of nursing and pharmacy workers working in hospitals relative to those working in all other sectors which is 0.93. We then apply the displacement wage scarring estimates Lachowska et al. (2020) to the worker’s out-of-market outside option. Specifically, we assume that having to find employment outside of the hospital market will lead to a displacement wage penalty of 40%, but our results are robust to penalties of 30% to 50%.⁴⁶ To get the final outside wage estimate, we deflate the mean nursing and pharmacy wage in hospitals operating in the HSA by both the non-hospital employee ratio (.93) and the displacement ratio (1-.4).

A.2.2 Colombian Manufacturing

For each Colombian manufacturing plant, we take the total wage bill for skilled workers and divide by the total number of skilled workers to get the average yearly earnings of a skilled worker. We assume that skilled workers forced to leave an industry (4-digit SIC code) would seek employment in a similar industry (a plant in the same 2-digit SIC code but different 4-digit SIC code).⁴⁷ We do not have an estimate of wage scarring effects for Colombian workers from this time period. Instead, we use a measure of displacement that would result in a period of unemployment when finding new employment. Eslava et al. (2010) find that Colombian workers experience a period of unemployment of, on average, 3.8 months following a plant closure. Using this information, we estimate the outside wage as the mean wage in the plants in the outside option reduced by 32% to reflect the expected time spent searching for new employment.

⁴⁶Earnings in another sector may also have a different utility value by revealed preference.

⁴⁷The 2-digit SIC labor market allows workers to consider jobs that are similar, but not identical to their current job, and avoids scenarios where there may only be one plant within a 4-digit SIC in a given region. For example, this means that a worker at a plant that makes “fertilizers and pesticides” may threaten to switch to plant that makes “paint varnish and lacquer” since both are chemicals industries, assuming these plants are in the same region. Wages are share-weighted.

B Simulated Mergers

B.1 U.S. Hospitals

We simulate all possible mergers between active, short-term, critical access, or children’s hospitals where we observe sufficient data and where mergers are possible ⁴⁸ within the local geographic market (Health Service Area) in 2006, resulting in a sample of 1392 potential mergers.⁴⁹ Our primary sample restriction is only including HSAs with multiple independently owned hospitals so that mergers are possible. This restriction results in a sample of 2114 hospitals located in 541 HSAs.

In order to calibrate our model and simulate mergers, we require that each hospital in an HSA report a market share and a wage data for nursing and pharmacy workers. Unfortunately, some data is often missing for at least some hospitals (especially very small hospitals) in HSAs with many hospitals. Because very small hospitals are unlikely to be important in price setting, we simply ignore hospitals in the merger simulations that are very small (accounting for less than 3% of patient discharges in an HSA) if those hospitals are missing on the data inputs critical to model calibration and merger simulation. If this data is missing for hospitals with greater than a 3% market share, we drop the HSA (and all of its hospitals) from the merger simulation exercise.

Our goal is to examine as many mergers as possible where the merging firms compete in either the labor and/or the product market. In many cases, an individual hospital is owned by a hospital system that owns many hospitals. Where there are multiple hospital systems operating in a geographic market, we assume that the acquiring system only purchases one (not all) of a competing systems hospitals in that market.⁵⁰ Information describing the HSA or HRR are mapped to hospitals in HCRIS by zip code using information from the

⁴⁸Our universe of hospitals consists of all hospitals appearing in the HCRIS data in 2006 that are active, short-term, critical access, or children’s hospitals.

⁴⁹We choose 2006 because it is the year that allows us to easily estimate hospital margins for each hospital system following the methodology developed by Dafny (2009) and Garmon (2017).

⁵⁰Simulating systems acquiring single hospitals rather than systems acquiring systems allows us to examine more potential mergers.

American Hospital Association Annual Survey⁵¹ When system information is missing, we assume hospitals are independent.

For within HSA mergers (scenario 1), we require that mergers increase HHI in the HSA, meaning that the acquiring system must have more nursing and pharmacy FTEs than the hospital system of the acquired hospital. This restriction, plus requiring HCRIS data on critical inputs, reduces our final sample to 1044 hospitals in 325 HSAs. For mergers within the HRR but across HSAs (scenario 2), we additionally impose the restriction that the merger must increase HHI in the HRR, meaning that the acquiring system has more nursing and pharmacy workers in the HRR than the acquired hospital. Furthermore, to avoid changes in the HHI within the HSA of the acquired hospital, we impose that the acquired hospital must be the only hospital in its system in the HSA, and that the acquiring system must not already be present in the HSA of the acquired system. These restrictions reduce the sample to 526 hospitals in 62 HRRs.

B.2 Colombian Manufacturing

Our goal for the tradable goods configuration is to simulate mergers with product market overlap but no labor market overlap. We achieve this by only simulating mergers between plants that have the same 4-digit SIC industry in Colombia, but operate in different geographic regions and therefore employ workers in separate labor markets.⁵² Therefore our main sample restriction is to keep only industries that have plants in multiple regions in order to simulate cross-region mergers. We then simulate all possible mergers between plants within four digit SIC industry and across region. We drop markets with $\lambda < .5$, affecting approximately 6% of our merger sample, due to issues with convergence of the model.

⁵¹See AHA Annual Survey data available at <https://www.ahadata.com/aha-annual-survey-database>. We also obtain system identifiers from AHA to establish pre-merger ownership structure in each market. When system is missing we impute that the hospital as independent. HSAs and HRR are obtained from Dartmouth's HSA and HRR crosswalk file available at <https://data.dartmouthatlas.org/supplemental/>.

⁵²Although the case in which tradable goods firms operate in the same labor market and sell into the same product market is of interest, for the purposes of illustrating our model, this case is similar to our first configuration but where concentration may not be exactly the same in the labor and product markets.

C Pass-through

In our US hospital and Colombian manufacturing simulations we do not see substantial pass-through of lower wages to lower prices. This occurs because affected worker wages are a very small share of firm revenue, and workers have relatively low bargaining power. To demonstrate that our model can generate substantial pass-through of worker wages to consumers and the firm, in this appendix we conduct merger simulations in markets with very different wages, margins, prices, and worker bargaining power than those in the main text. We stress that these are artificial markets; that is, they do not correspond to any specific industry. The findings shown here demonstrate that under our model, mergers can result in significant transfers from workers to consumers in some types of markets. However, whether mergers take place in these types of markets is an open question.

To construct these markets, we modify the hospital and Colombian manufacturer data using the following algorithm. First, we impose that workers and employers have equal bargaining power ($\lambda = 0.5$). This change results in a change in the equilibrium level of wages, prices, and the firms' margins. The lower λ implies that workers earn a greater share of rents. Relative to the markets discussed in the text, these alternative markets have much higher wages and prices and lower margins. In particular, median nursing and pharmacist worker wages in this alternative economy are much higher (around \$3 million versus \$65,000 in our observed markets), and in the modified Colombian manufacturing application wages are about \$21,000 (versus \$6,000 in the observed markets). Similarly, in the modified hospital markets, prices are higher, \$9 million versus \$6 million in the observed markets. Summary statistics describing these alternative markets are shown in Table 5. All other model inputs and outputs, excluding λ , which is imposed to be 0.5, are identical to those presented in Tables 1-3 in the main text.

Having created alternative markets, we now simulate mergers using the same approach used in the main text.⁵³ Figure 4 presents the the results from our full model. For con-

⁵³Note that while pre-merger wages, prices, margins worker (firm) bargaining power are different in the

Table 5: Pre-Merger Summary Statistics that Differ when $\Lambda = 0.5$

	50%	5%	25%	75%	95%
Product and Labor Overlap					
Hospital Wage (\$100k/year)	31	7	17	55	117
Hospital Margin per Worker (\$100k/year)	23	4.7	12	43	91
Hospital Price per Worker (\$100k/year)	91	37	63	142	280
Only Labor Overlap					
Hospital Wage (\$100k/year)	30	9.4	19	54	105
Hospital Margin per Worker (\$100k/year)	21	4.4	11	38	77
Hospital Price per Worker (\$100k/year)	85	36	64	135	247
Only Product Overlap					
Plant Wage (\$100k/year)	0.21	0.084	0.13	0.46	1.1
Plant Margin per Worker (\$100k/year)	0.15	0.04	0.096	0.41	0.89
Plant Price per Worker (\$100k/year)	1.2	0.34	0.68	2.2	4.5

venience, we repeat the results from the models estimated using market data identical to those shown in Figure 2 in green (left), and the new results from modified markets in orange (right). In the alternative markets, we find that the percent reduction in worker surplus resulting from mergers is smaller. That is, having higher bargaining power tends to reduce the proportional drop in rents captured by workers resulting from mergers. However, because workers collectively earn a larger portion of the surplus in the pre-merger equilibrium (accounting for around 50% of firm costs in hospital applications and 32% in the manufacturing application, as opposed to 2% and 8% respectively, in the original results), the reduction in worker surplus from mergers results in a meaningful transfer from workers to consumers and firms. For example, in the second case where the simulated mergers affect the labor market but not the product market, consumer and firm surplus increases proportionately by much more in the alternative markets. In fact, in all three cases, consumer harm from mergers is lower in the alternative markets because the reduced worker payroll is partially passed through to reduce (or even offset) the change in final goods prices.

alternative simulations, the mergers generate the same changes in market structure (ΔHHI) as the original simulations.

Comparison of Welfare Effects Across Different Model Configurations and Specifications

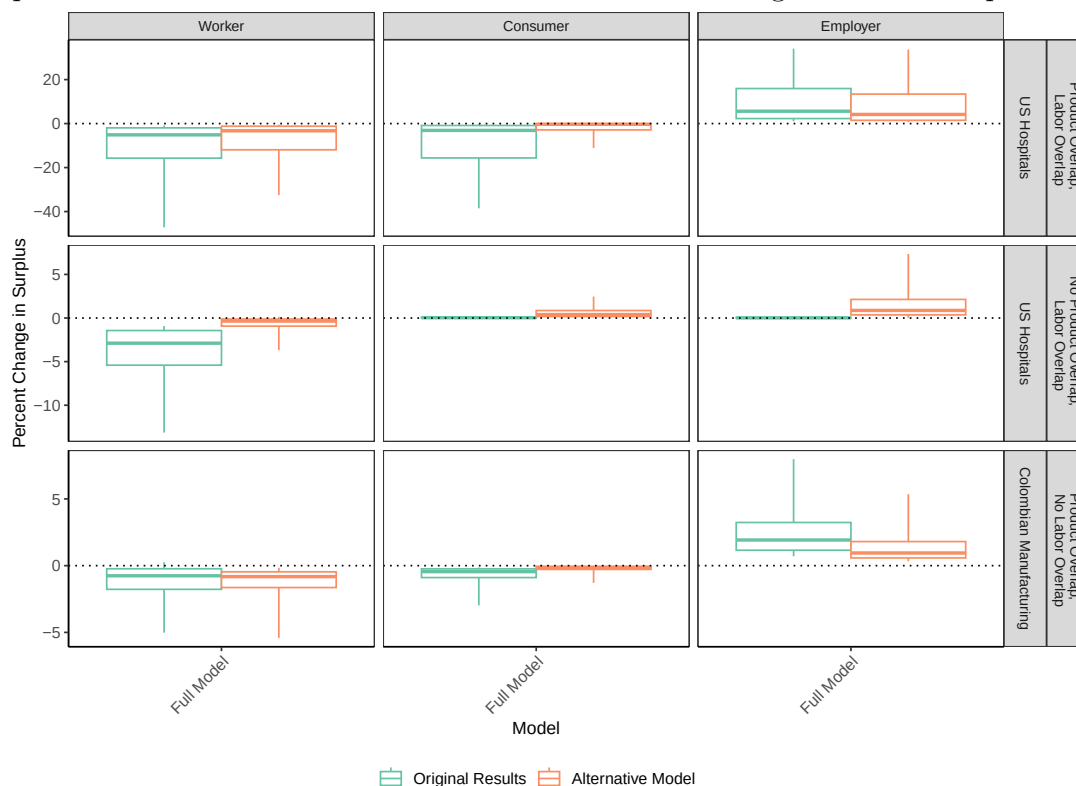


Figure 4 displays worker, consumer, and employer surplus effects of the mergers in percentage terms, including the effect of changes to shares. Each box and whisker plot shows the 5th, 25th, 50th, 75th, and 95th percentiles of outcomes among the simulated mergers. We show the results of each model either with the λ implied by observed margins (left, green) and wages, or with a λ imposed to be .5 (right, orange), and wages and margins adjusted accordingly in the pre-merger equilibrium.